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Assignment - 05

Module - 05

Problem Solving with AI (Claude Code + JavaScript)

Q1. JavaScript Coding Practice

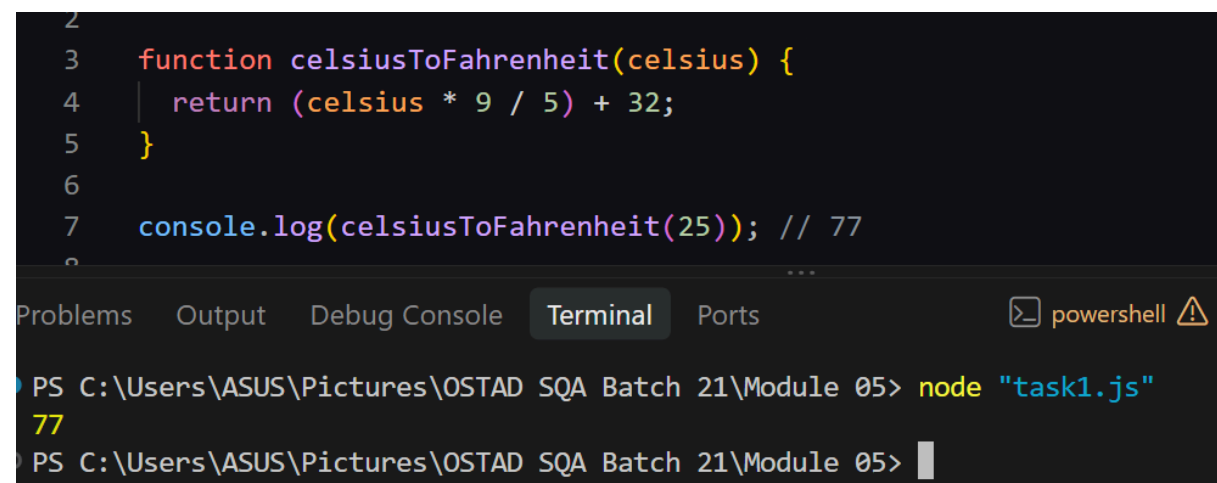
Answer:

i) Convert Celsius to Fahrenheit

Code:

```
function celsiusToFahrenheit(celsius) {  
    return (celsius * 9 / 5) + 32;  
}  
  
console.log(celsiusToFahrenheit(25)); // 77
```

Screenshot:



The screenshot shows a code editor with a dark theme. The code is as follows:

```
2  
3 function celsiusToFahrenheit(celsius) {  
4     return (celsius * 9 / 5) + 32;  
5 }  
6  
7 console.log(celsiusToFahrenheit(25)); // 77  
8
```

Below the code editor, there is a terminal window. The terminal shows the command `node "task1.js"` being executed, and the output `77` is displayed. The terminal path is `PS C:\Users\ASUS\Pictures\OSTAD SQA Batch 21\Module 05>`.

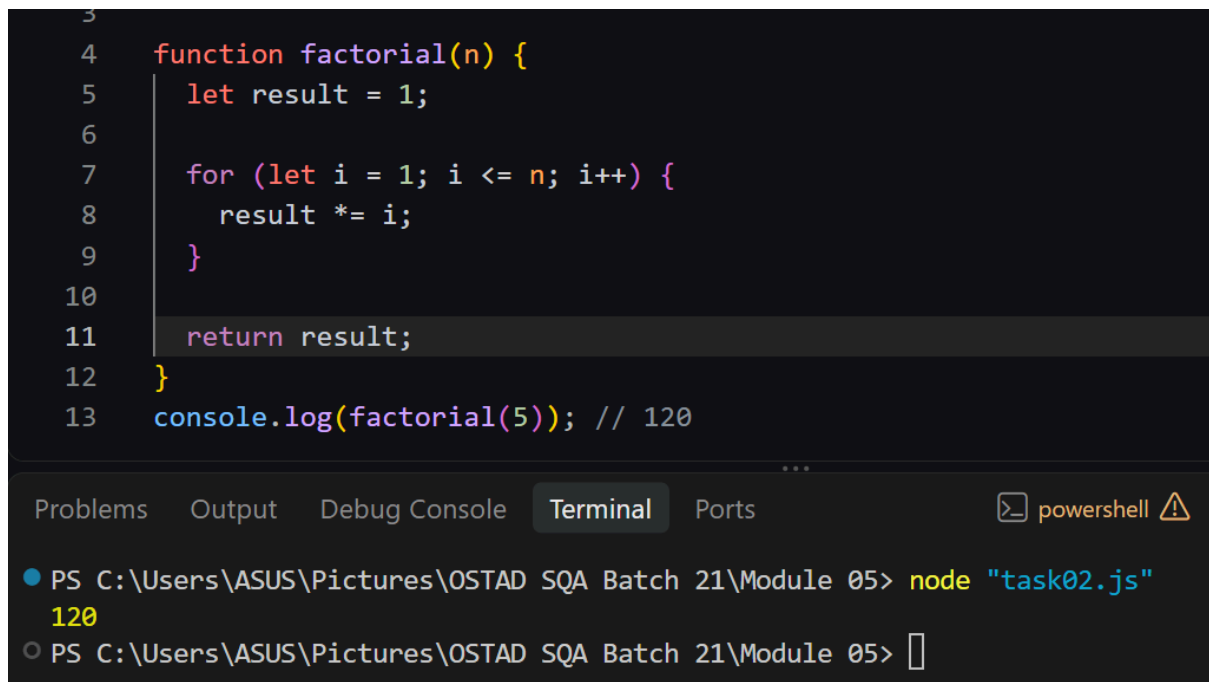
Explanation: The function takes a temperature in Celsius as input and converts it to Fahrenheit using the formula $F = (C \times 9/5) + 32$. It then returns the calculated Fahrenheit temperature.

ii). Find the Factorial of a Number

Code:

```
function factorial(n) {  
  let result = 1;  
  
  for (let i = 1; i <= n; i++) {  
    result *= i;  
  }  
  
  return result;  
}  
console.log(factorial(5)); // 120
```

Screenshot:



The screenshot shows a code editor with the following code:

```
3  
4  function factorial(n) {  
5      let result = 1;  
6  
7      for (let i = 1; i <= n; i++) {  
8          result *= i;  
9      }  
10  
11     return result;  
12 }  
13 console.log(factorial(5)); // 120
```

Below the code editor is a terminal window with the following output:

```
PS C:\Users\ASUS\Pictures\OSTAD SQA Batch 21\Module 05> node "task02.js"  
120  
PS C:\Users\ASUS\Pictures\OSTAD SQA Batch 21\Module 05> 
```

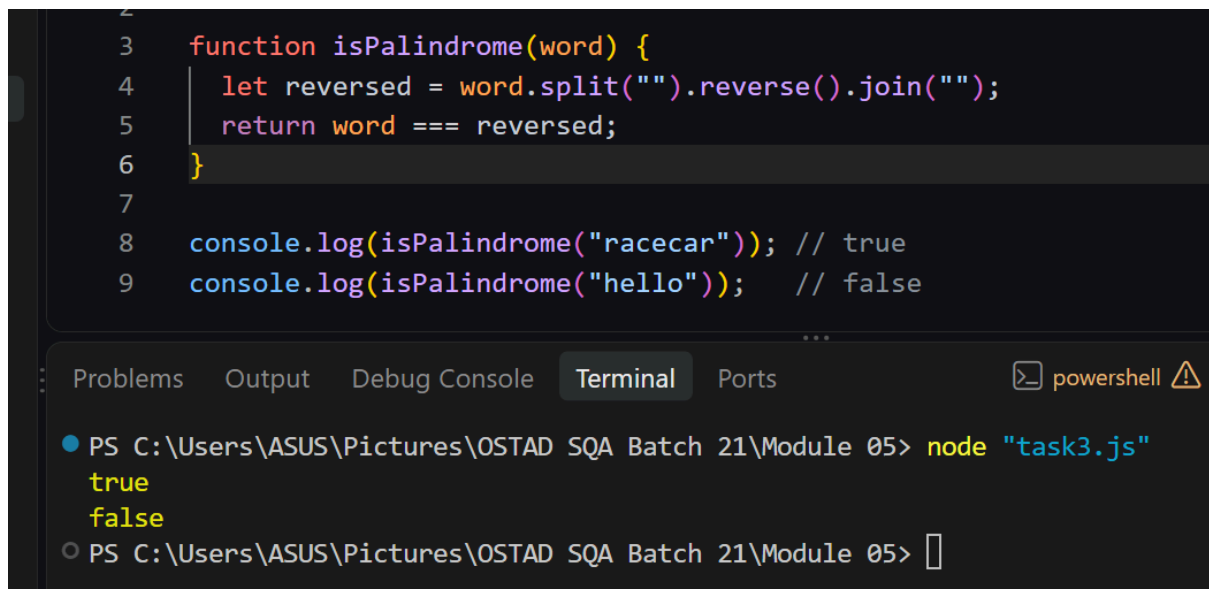
Explanation: The function starts with $\text{result} = 1$ and multiplies it by every number from 1 to the given number. For example, $5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$. Finally, the function returns the factorial result.

iii) Check for Palindrome

Code:

```
function isPalindrome(word) {  
  let reversed = word.split("").reverse().join("");  
  return word === reversed;  
}  
  
console.log(isPalindrome("racecar")); // true  
console.log(isPalindrome("hello"));  // false
```

Screenshot:



The screenshot displays a code editor with the following JavaScript code:

```
2  
3  function isPalindrome(word) {  
4    let reversed = word.split("").reverse().join("");  
5    return word === reversed;  
6  }  
7  
8  console.log(isPalindrome("racecar")); // true  
9  console.log(isPalindrome("hello"));  // false
```

Below the code editor, the terminal window shows the output of running the code:

```
PS C:\Users\ASUS\Pictures\OSTAD SQA Batch 21\Module 05> node "task3.js"  
true  
false  
PS C:\Users\ASUS\Pictures\OSTAD SQA Batch 21\Module 05> 
```

Explanation: The function reverses the given word and compares it with the original word. If both are the same, the word is a palindrome and the function returns true; otherwise, it returns false.

iv) Sum of Array Elements

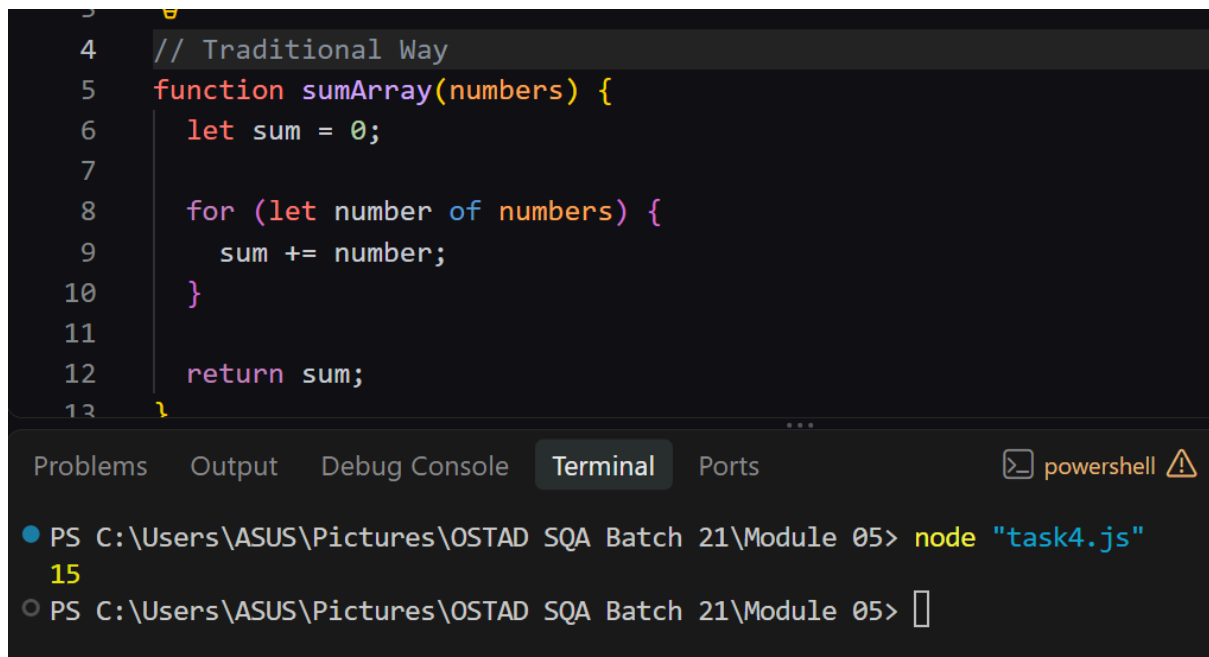
Code:

```
// Traditional Way
function sumArray(numbers) {
  let sum = 0;

  for (let number of numbers) {
    sum += number;
  }

  return sum;
}
console.log(sumArray([1, 2, 3, 4, 5])); // 15
```

Screenshot:



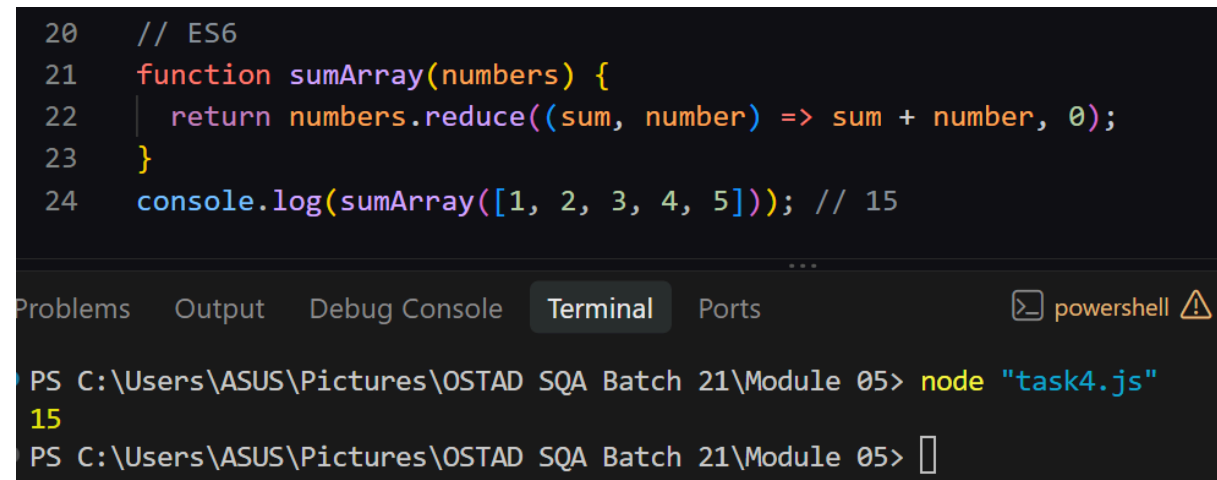
The screenshot shows a code editor with a dark theme. The code for the `sumArray` function is visible, starting from line 4. The function initializes `sum = 0`, loops through each element in the `numbers` array, adds it to the sum, and returns the final sum. Below the code editor, a terminal window is open, showing the command `node "task4.js"` being executed. The output of the command is `15`.

Explanation: The function starts with `sum = 0` and loops through each number in the array. It adds each number to the sum and finally returns the total.

Code:

```
// ES6
function sumArray(numbers) {
  return numbers.reduce((sum, number) => sum + number, 0);
}
console.log(sumArray([1, 2, 3, 4, 5])); // 15
```

Screenshot:



The screenshot shows a code editor with the following code:

```
20 // ES6
21 function sumArray(numbers) {
22 |   return numbers.reduce((sum, number) => sum + number, 0);
23 | }
24 console.log(sumArray([1, 2, 3, 4, 5])); // 15
```

Below the code editor is a terminal window with tabs for Problems, Output, Debug Console, Terminal, and Ports. The Terminal tab is active, showing a PowerShell prompt. The command `node "task4.js"` has been executed, and the output `15` is displayed.

Explanation: The `reduce()` method goes through each element of the array and adds it to the accumulated sum. The `0` is the initial value of the sum. Finally, it returns the total sum of all elements.

v) FizzBuzz

Problem Statement: Write a loop from 1 to 15 that prints "Fizz" for numbers divisible by 3, "Buzz" for numbers divisible by 5, "FizzBuzz" for numbers divisible by both, and the number itself otherwise.

Code:

```
for (let i = 1; i <= 15; i++) {  
  
    if (i % 3 === 0 && i % 5 === 0) {  
  
        console.log("FizzBuzz");  
  
    } else if (i % 3 === 0) {  
  
        console.log("Fizz");  
  
    } else if (i % 5 === 0) {  
  
        console.log("Buzz");  
  
    } else {  
  
        console.log(i);  
  
    }  
  
}
```

Explanation: The loop runs from 1 to 15. The % operator checks whether a number is divisible by 3 or 5. We check for both first, so numbers divisible by both print "FizzBuzz". Otherwise, multiples of 3 print "Fizz", multiples of 5 print "Buzz", and all other numbers print themselves.